Register Number:

Name of the Candidate:

3219

B.E. DEGREE EXAMINATION, 2022

(COMMON TO CS, IT AND AI & ML)

(FOURTH SEMESTER)

CSBS-401/ITBS401/AIBS-401: DISCRETE MATHEMATICS

(for those who joined from 2018-2019)

December] [Time : 3 Hours

Maximum : 75 Marks

SECTION – A

Answer ALL questions **(10 × 1 =10)**

1. *Define Tautological Implication.* *1CO1K1*
2. *State Absorption laws in logic.* *1CO1K1*
3. *If A={1} and B={1,2} find A×B.* *1CO2K1*
4. *Define subset of a set. What is mean by proper subset?* *1CO2K1*
5. *State the modular inequality of a Lattice.* *1CO3K1*
6. *Define Boolean algebra.* *1CO3K1*
7. *Define a Monoid with an example* *1CO4K1*
8. *what is the identity element of the algebraic system* $\{s,*\}$ *Where S is the set of integers and * is defined by* $a * b = a + b + 2$ *for all* $a, b \in S$. *1CO4K1*
9. *Define regular graph with an example* *1CO5K1*
10. *Define graph isomorphism.* *1 CO5 K1*

SECTION – B

Answer either (a) or (b) in each Question **(5 × 3 =15)**

11 (a) *Prove that* $P \to (Q \to R) \Leftrightarrow (P \wedge Q) \to R$ *3CO1K2*

OR

(b) *Prove that* $\neg Q \wedge (P \to Q) \Rightarrow \neg P$ *3CO1K2*

12 (a) *If A,B and C are any three sets Prove that* $A - (B \cup C) = (A - B) \cap (A - C)$ *3CO2K2*

OR

(b) *If R is the relation on* $A = \{1,2,3\}$ *such that* $(a, b) \in R$, *if and only if* $a + b = even\ number$ *Find the relational matrix* M_R *also find the relational matrix of* R^{-1} *and* R^2 *3CO2K2*

13 (a) Draw the Hasse diagram for the divisibility relation on the set D_{24} 3CO3K2

OR

(b) State and prove the isotonic property of a Lattice. 3CO3K2

14 (a) show that in a group G, $(a*b)^{-1} = b^{-1} * a^{-1}, for\ all\ a,b \in G$ 3CO4K2

OR

(b) Show that $\{(1,2,3,4,5,6),\times_7\}$ *is a cyclic group.* What are its generators? 3CO4K2

15 (a) Explain konisberg bridge problem. Represent the problem by means of graph. Does the problem have a solution? 3CO5K2

OR

(b) Define Bipartite graph. Draw $K_{2,3}$ *and* $K_{3,3}$ graphs. 3CO5K2

SECTION – C

Answer either (a) or (b) in each Question **(4 × 10 =40)**

16 (a) Obtain the Principal Conjunctive normal form of $(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$ 10 CO1 K3

OR

(b) Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q 10 CO1 K3

17 (a) show tht the relation R "*congruent modulo* m" is an equivalence relation on the set of integers. Where m is a positive integer greater than 1. 10 CO2 K3

OR

(b) Prove that $\oplus$ is an associative operation. 10 CO2 K3

18 (a) Show that in a distributive lattice $\{L,\vee,\wedge\}$ if an element $a \in L$ has a complement then it is unique. 10 CO3 K3

OR

(b) Simplify the Boolean function $f(a,b,c,d) = \Sigma(0,1,2,3,4,6,7,8,9,11,15)$ 10 CO3 K3

19 (a) show that $\{s_3,*\}$, where $s = \{1,2,3\}$ is a group under the operation of right composition of permutation. 10 CO4 K3

OR

(b) List the code words generated by the encoding function $e: B^3 \rightarrow B^6$ whose parity check matrix is $H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ 10 CO4 K3

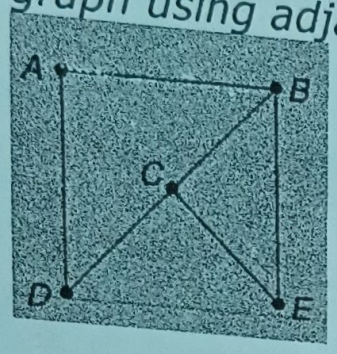
3219

SECTION – D

Answer either (a) or (b)

(1 × 10 =10)

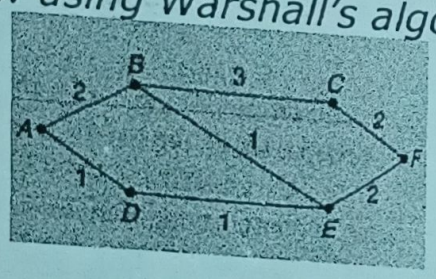
20(a) Summarize all the paths of length 4 between D and E for the following graph using adjacency matrix



OR

10 CO5 K4

(b)Evaluate the shortest distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the weighted graph given below using Warshall's algorithm.



10CO5K4
